PL Assignment III

September 2, 2025

Q1. Prove the following:

1. (Hypothetical Syllogism, HS) $\{\alpha \to \beta, \beta \to \gamma\} \vdash \alpha \to \gamma$.
2. $\neg\alpha \to (\alpha \to \beta)$.
3. If $\Gamma \vdash \Box$, then $\Gamma \vdash \alpha$, for every wff α.
4. (Reductio ad absurdum, RAA) If $\Gamma \cup \{\neg\alpha\} \vdash \Box$, then $\Gamma \vdash \alpha$.
5. $\vdash (\neg\alpha \to \alpha) \to \alpha$.
6. $\vdash \neg\neg\alpha \to \alpha$.
7. $\vdash (\alpha \to \neg\alpha) \to \neg\alpha$.
8. If $\Gamma \cup \{\alpha\} \vdash \Box$, then $\Gamma \vdash \neg\alpha$.
9. $\vdash \alpha \to \neg\neg\alpha$.
10. $\vdash (\alpha \to \beta) \to (\neg\beta \to \neg\alpha)$.
11. $\vdash (\beta \to \alpha) \to ((\neg\beta \to \alpha) \to \alpha)$.
12. If $\Gamma \cup \{\beta\} \vdash \alpha$ and $\Gamma \cup \{\neg\beta\} \vdash \alpha$ then $\Gamma \vdash \alpha$.
13. - $\{\neg(\alpha \to \neg\beta)\} \vdash \alpha$.
    - $\{\neg(\alpha \to \neg\beta)\} \vdash \beta$.
14. - $\{\alpha\} \vdash \neg\alpha \to \beta$.
    - $\{\beta\} \vdash \neg\alpha \to \beta$.

Q2. Prove the following:

1. $\vdash \alpha \to (\alpha \vee \beta)$.
2. $\vdash \neg(\alpha \wedge \neg\alpha)$.
3. $\vdash \alpha \to \neg(\neg\alpha \wedge \beta)$.
4. $\vdash (\alpha \vee \beta) \vee (\beta \to \alpha)$.
5. $\{\alpha, \neg\beta\} \vdash \neg(\neg\alpha \vee \beta)$.
6. $\{\neg(\alpha \vee \beta)\} \vdash \neg\alpha$.
7. $\{\alpha \to \beta, \gamma \vee \alpha\} \vdash \gamma \vee \beta$.
8. $\{\alpha \to (\beta \wedge \gamma)\} \vdash \alpha \to \beta$.
9. $\{(\alpha \vee \beta) \to \gamma\} \vdash \alpha \to \gamma$.
10. $\{\alpha \to \gamma\} \vdash (\alpha \wedge \beta) \to \gamma$.

Q3. Let Γ and Δ be sets of formulae. Prove the following:

1. If $\Gamma \subseteq \Delta$ and $\Gamma \vdash \alpha$, then $\Delta \vdash \alpha$.
2. If $\Gamma \cup \{\alpha\} \vdash \beta$ and $\Gamma \vdash \alpha$, then $\Gamma \vdash \beta$.
3. If $\Gamma \vdash (\alpha \to \beta)$, then $\Gamma \cup \{\alpha\} \vdash \beta$.
4. If $\{\alpha_1, \alpha_2, \ldots, \alpha_n\} \vdash \beta$ and $\vdash \alpha_1, \ldots, \vdash \alpha_n$, then $\vdash \beta$.
5. If $\Delta \vdash \alpha$ and $\Gamma \vdash \beta$ for every $\beta \in \delta$, then $\Gamma \vdash \alpha$.

Q4. Let $\Gamma := \{p_1, \neg p_2 \vee p_3\}$. Show that $\Gamma \nvdash p_3$.

Q5. Let $\Gamma := \{p_n : n \geq 2\}$. Prove the following:

1. $\Gamma \nvdash p_1$.
2. $\Gamma \nvdash \neg p_1$.

Q6. Let $\Gamma \cup \{\neg\alpha\}$ be satisfiable. Show that $\Gamma \nvdash \alpha$.

Q7. Use the Soundness theorem to show that the following argument forms are valid:

1. $\alpha \vee \beta, \gamma \vee \delta, \neg\alpha \vee \neg\gamma, \neg\beta \vee \neg\delta, \beta \vee \gamma \therefore \neg\alpha \wedge (\beta \wedge (\gamma \wedge \neg\delta))$.
2. $\alpha \vee (\beta \vee \gamma), \alpha \vee \neg\beta, \neg\alpha \vee \delta, \gamma \vee \neg\delta \therefore \gamma$.
3. $\alpha \vee (\beta \vee \gamma), \neg\alpha \vee \beta, \alpha \vee \neg\gamma, \neg\beta \vee \gamma \therefore \alpha \wedge (\beta \wedge \gamma)$.
4. $\alpha \vee \beta, \beta \to (\alpha \wedge \gamma) \therefore \alpha$.

Q8. For each of the following, decide whether Γ is consistent or inconsistent. To show that Γ is consistent, find a valuation that satisfies Γ. To show that Γ is inconsistent, either show that Γ is unsatisfiable or find some wff α such that both $\Gamma \vdash \alpha$ and $\Gamma \vdash \neg\alpha$. p_i's are propositional variables.

1. $\Gamma := \{p_1, p_2, \neg(p_2 \vee p_3)\}$.
2. $\Gamma := \{p_1 \vee \neg p_2, \neg(p_1 \vee p_2)\}$.
3. $\Gamma := \{p_1 \vee \neg p_2, \neg(p_1 \vee p_2), p_1 \to p_2\}$.
4. $\Gamma := \{\neg(p_1 \vee p_2), \neg p_2 \vee p_3, (p_2 \vee p_3) \to p_4\}$.

Q9. Let Γ be the set of all wffs of the form $\neg\alpha \vee \beta$. Is Γ consistent?

Q10. Let Γ be a consistent set of wffs. Show that the following are equivalent:

1. $\alpha \in \Gamma$ or $\neg\alpha \in \Gamma$ for every wff α.
2. If $\Gamma \cup \{\alpha\}$ is consistent, then $\alpha \in \Gamma$.

Q11. Let Γ be maximally consistent, and α, β any wffs. Prove the following.

1. $\Gamma \vdash \alpha$, if and only if $\alpha \in \Gamma$.
2. $\neg\alpha \in \Gamma$ if and only if $\alpha \notin \Gamma$.
3. $\alpha \to \beta \in \Gamma$ if and only if $\alpha \notin \Gamma$, or $\beta \in \Gamma$.
4. $\alpha \wedge \beta \in \Gamma$ if and only if $\alpha \in \Gamma$ and $\beta \in \Gamma$.
5. $\alpha \vee \beta \in \Gamma$ if and only if $\alpha \in \Gamma$ or $\beta \in \Gamma$.

Q12. Let Γ be a consistent set of wffs. Show that the following are equivalent:

1. For every wff α, $\Gamma \vdash \alpha$ or $\Gamma \vdash \neg\alpha$.
2. There is exactly one valuation that satisfies Γ.